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# Exercise Sheet 2 for Advanced Algorithms (Winter 2026/27)

***Hand In:*** *Until 2026-10-30 18:00, on ILIAS.*

## Problem 1

30 + 30 + 30 points

For each of the following problems, either sketch a pseudopolynomial algorithm for solving the problem (proving that the problem is only weakly NP-hard), or show that the problem is strongly NP-hard.

For integer-input problems with numbers in $\mathbb{Z}$, MaxInt of an instance is the largest *absolute value* occurring in the input.

a) Partition

**Given:** A sequence of integers $x_1, \dots, x_n \in \mathbb{N}$.

**Question:** $\exists I \subseteq [n] \;:\; \sum_{i \in I} x_i = \sum_{i \in [n] \setminus I} x_i$ ?

b) Subset Sum

**Given:** $x_1, \dots, x_n \in \mathbb{Z}$.

**Question:** $\exists I \subseteq [n] \;:\; I \neq \emptyset \wedge \sum_{i \in I} x_i = 0$ ?

c) 0/1 Integer Programming

**Given:** Integer linear program (ILP): $A \in \mathbb{Z}^{m \times n}$, $b \in \mathbb{Z}^m$ and $c \in \mathbb{Z}^n$ and $k \in \mathbb{Z}$

**Question:** Is there $x \in \{0,1\}^n$ with $Ax \leq b$ and $c^T x \geq k$ ?